

Object Oriented Programming for Microcontrollers

Engineering Technology

May, 2026



Object Oriented Programming for Microcontrollers (A11889)

Short Title:	Object Oriented Programming	
Faculty	Science and Computing	
Department	Engineering Technology	
Cluster	Software and Web Development	
Author	CLAIRE.FITZPATRICK	
Credits	5	Level: Intermediate

Description of Module / Aims

This module follows on from the module called Advanced Programming Concepts and builds on concepts introduced there. Object Oriented Programming module teaches the learner about advanced object oriented concepts such as classes, inheritance and polymorphism.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (WD_EAUTO_B)	2 / 4 / M
PROG-0081	BEng (Hons) in Electronic Engineering (WD_EONIC_B)	2 / 4 / M
	BEng (Hons) in Information Engineering (International) (WD_EEELC_BI)	2 / 4 / M

Indicative Content

- Object Oriented Programming
- C++ Classes and Constructors
- Abstract Data Types (ADTs)
- Overloaded Operators
- Try Catch Blocks (Error Processing)
- Inheritance and Polymorphism

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe approaches to programming and problem solving.
2. Write, debug and test well-styled and appropriately-documented programs.
3. Demonstrate the use of OO techniques, using knowledge of the concepts when writing code and when using class libraries.
4. Develop applications to control sensor inputs and outputs connected to a microcontroller.
5. Manipulate microcontroller class libraries to control sensors and display outputs on various display devices.

Learning and Teaching Methods

- This course will be presented by a combination of lectures and computer-based practicals whilst capitalising on a web-enhanced learning environment
- The lectures will be used to introduce new topics and their related concepts
- The emphasis on course delivery will be hands-on, problem solving both individually and in small class groups using problem based worksheets during the practical sessions
- Self-directed learning will be encouraged throughout the duration of the module

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>Practical</i>	80%	1,2,3,4,5
<i>In-Class Assessment</i>	20%	1,2,3

Assessment Criteria

70% -

100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Deitel, P. and H. Deitel. *_C++ How to Program (Early Objects Version)_*. 9th. New Jersey: Pearson Education, 2014.
- Savitch, W. *_Problem Solving with C++_*. 9th. New Jersey: Pearson Education, 2015.

Requested Resources

- Room Type: Computer Lab